No limit cycle in the Banaji–Boros–Hofbauer three-reaction example

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Dulac argument and subfamily: Claude Opus 5.5 exploration agents (scout and verifier, effort high), 17 Oct 2026 · Certificate and manuscript: Claude Opus 5.5 (effort xhigh) · Verification: Claude Opus 5.5 (effort xhigh), fresh context, 18 Oct 2026 (confirmed with corrections)

Abstract. Banaji, Boros and Hofbauer classified the three-reaction planar quadratic mass-action networks that admit an Andronov–Hopf bifurcation, and asked whether the networks of their cases 9 and 10 without such a bifurcation can nevertheless have a limit cycle; as an example they asked about the network $2X \to 3X + 2Y,\ X + Y \to 0,\ Y \to X + Y$. We show that this system has no periodic orbit for any rate constants, using a Dulac function that turns the divergence into a negative definite quadratic form. The same argument shows that in the subfamily $2X \to (2 + c_1)X + d_1Y,\ X + Y \to 0,\ Y \to c_3X + Y$, periodic orbits can only occur in the Andronov–Hopf region $c_1 < d_1 < 2c_1$.

The question

Banaji, Boros and Hofbauer [1] classify the three-reaction, planar, quadratic mass-action systems that admit an Andronov–Hopf bifurcation (their Theorem 9, cases 9 and 10). Throughout, theorem numbers refer to the arXiv version (arXiv:2304.02303v1); in the published version this classification is Theorem 3, with the same case numbering and the same notation. In their Conclusions (§6, identical in both versions) they write:

> *"However, we leave it open whether the networks in cases 9 and 10 admitting no Andronov–Hopf bifurcations can have a limit cycle. For example, we do not know whether the system $2X \to 3X + 2Y$ (κ_1), $X + Y \to 0$ (κ_2), $Y \to X + Y$ (κ_3), $\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y$, $\dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$ admits a limit cycle."*

We settle the example (it has no periodic orbit at all, for any rate constants) and, with the same argument, the subfamily of case-9 networks to which it belongs.

Theorem 1. For all $\kappa_1, \kappa_2, \kappa_3 > 0$, the system

$$\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y, \qquad \dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$$

has no periodic orbit in the closed positive quadrant. In particular it has no limit cycle.

Theorem 2. Let c_1, c_3, d_1 be positive integers and consider the networks

$$2X \xrightarrow{\kappa_1} (2 + c_1)X + d_1Y, \qquad X + Y \xrightarrow{\kappa_2} 0, \qquad Y \xrightarrow{\kappa_3} c_3X + Y,$$

i.e. case 9 of Theorem 9 [1] with, in the notation of (13) [1], $c = (c_1, c_2, c_3) = (c_1, -1, c_3)$ and $d = (d_1, d_2, d_3) = (d_1, -1, 0)$ (the X- and Y-components of the three

reaction vectors). The mass-action system

$$\dot{x} = c_1\kappa_1 x^2 - \kappa_2 xy + c_3\kappa_3 y, \qquad \dot{y} = d_1\kappa_1 x^2 - \kappa_2 xy$$

has a periodic orbit in the closed positive quadrant for some κ only if $c_1 < d_1 < 2c_1$, which is exactly the region where it undergoes an Andronov–Hopf bifurcation Theorem 9 [1]. For $d_1 \le c_1$ or $d_1 \ge 2c_1$ it has no periodic orbit in $\mathbb{R}^2_{\ge 0}$ for any $\kappa > 0$.

The example of Theorem 1 is the case $c_1 = c_3 = 1$, $d_1 = 2$, on the boundary $d_1 = 2c_1$ of the Hopf region.

1 Proofs

Proof (of Theorem 2, case $d_1 \ge 2c_1$)**.** Write $F = (f, g)$ for the vector field and take the Dulac function $B(x, y) = y^{-b}$ on $\mathbb{R}^2_{>0}$, with

$$b = 1 + \frac{2\kappa_1(d_1 - c_1)}{\kappa_2}.$$

A direct computation gives

$$y^{b+1}\,\mathrm{div}(BF) = Q(x, y) := \frac{d_1\kappa_1\big(2(c_1 - d_1)\kappa_1 - \kappa_2\big)}{\kappa_2}\,x^2 + 2d_1\kappa_1\,xy - \kappa_2\,y^2,$$

a quadratic form with y^2 coefficient $-\kappa_2 < 0$ and discriminant

$$\Delta = 4d_1\kappa_1(\kappa_1(2c_1 - d_1) - \kappa_2).$$

(The exponent b minimises Δ among Dulac functions y^{-b}.) If $d_1 \ge 2c_1$, then $\kappa_1(2c_1 - d_1) - \kappa_2 \le -\kappa_2 < 0$, so $\Delta < 0$ and Q is negative definite: $\mathrm{div}(BF) < 0$ on the open quadrant. Since $\mathbb{R}^2_{>0}$ is simply connected and B is smooth on it, the Bendixson–Dulac criterion excludes periodic orbits contained in it.

On $\{y = 0,\ x > 0\}$ we have $\dot{y} = d_1\kappa_1 x^2 > 0$, on $\{x = 0,\ y > 0\}$ we have $\dot{x} = c_3\kappa_3 y > 0$, and the origin is an equilibrium. A periodic orbit cannot contain an equilibrium. If a solution γ of a periodic orbit in the closed quadrant passed through a point $p = \gamma(t_0)$ with $p = (x_0, 0)$, $x_0 > 0$, then $y(t) = \dot{y}(t_0)(t - t_0) + o(t - t_0) < 0$ for $t < t_0$ close to t_0, so the orbit would leave the closed quadrant; the same holds on the positive y-axis. Hence every periodic orbit in the closed quadrant would lie in the open quadrant, which is impossible. □

Proof (of Theorem 2, case $d_1 \le c_1$)**.** The stoichiometric matrix $\Gamma = \begin{pmatrix} c_1 & -1 & c_3 \\ d_1 & -1 & 0 \end{pmatrix}$ has rank 2 (its rows are not proportional because $c_3 \ne 0$), so $\ker\Gamma$ is spanned by $u = c \times d = (c_3,\ c_3 d_1,\ d_1 - c_1)$. A positive equilibrium x requires a positive flux vector $v(x) \in \ker\Gamma \cap \mathbb{R}^3_{>0}$, i.e. u or $-u$ positive, which fails when $d_1 \le c_1$. So there is no positive equilibrium. A periodic orbit in the open quadrant would enclose an equilibrium (index theory), which would be positive; hence there is none, and the boundary argument above excludes orbits touching the axes. □

Proof (of Theorem 1)**.** This is $c_1 = c_3 = 1$, $d_1 = 2$: $b = 1 + 2\kappa_1/\kappa_2$, $Q = -\frac{2\kappa_1(2\kappa_1 + \kappa_2)}{\kappa_2}\,x^2 +$

$4\kappa_1 xy - \kappa_2 y^2$ and $\Delta = -8\kappa_1\kappa_2 < 0$. □

Remark 3. In the example, the unique positive equilibrium is $(2\kappa_3/\kappa_2,\ 4\kappa_1\kappa_3/\kappa_2^2)$, with $\det J = 4\kappa_1\kappa_3^2/\kappa_2 > 0$ and $\operatorname{tr} J = -2\kappa_3 < 0$.

For case 9 with $c_3 > 0$, Theorem 9 [1] gives an Andronov–Hopf bifurcation iff $c_1 > 0$, $c_2 = -1$, $d_1 > 0$, $d_2 \geq -1$, $d_3 \geq -1$ (all satisfied in the subfamily) and $\frac{1}{2}\,(d_3/c_3 + d_1/c_1) < d_2/c_2 < d_1/c_1$. In the subfamily, $d_2/c_2 = 1$ and $d_3 = 0$, so the condition reads $c_1 < d_1 < 2c_1$; for the example, $\frac{1}{2}\,(0+2) = 1 < 1 < 2$ fails by equality. Theorem 2 shows that in this subfamily "no Andronov–Hopf bifurcation" implies "no periodic orbit". ⌟

Remark 4 (added at verification)**.** (i) *Sharpness.* For $c_1 < d_1 < 2c_1$ the positive equilibrium satisfies $y^*/x^* = d_1\kappa_1/\kappa_2$, and $\operatorname{tr} J$ has the sign of $\kappa_1(2c_1 - d_1) - \kappa_2$, while $\det J > 0$. So the Hopf curve $\operatorname{tr} J = 0$ is exactly the curve $\Delta = 0$. On it, $Q = -\kappa_2\,(y - rx)^2$ with $r = d_1/(2c_1 - d_1)$: $\operatorname{div}(BF) \leq 0$ vanishes only on a ray, and the Bendixson–Dulac argument (Green's formula) still applies. Hence, in the whole subfamily, there is no periodic orbit in $\mathbb{R}^2_{\geq 0}$ whenever $\kappa_2 \geq \kappa_1(2c_1 - d_1)$, i.e. whenever $\operatorname{tr} J \leq 0$ at the positive equilibrium (if there is one). For $c_1 < d_1 < 2c_1$ and $\kappa_2 < \kappa_1(2c_1 - d_1)$ the equilibrium is a repellor, and limit cycles do occur near the supercritical Hopf curve Theorem 9 [1].

(ii) *Other values of* c_1. The proof of the case $d_1 \geq 2c_1$ uses neither $c_1 \geq 1$ nor integrality. For the admissible values $c_1 \in \{-2, -1, 0\}$ (target $(2 + c_1)X + d_1 Y$) with $d_1 \geq 1$, the same Dulac function gives $\Delta = -4d_1\kappa_1(\kappa_2 + \kappa_1(d_1 - 2c_1)) < 0$, so these networks have no periodic orbit either. These networks are also in case 9 without an Andronov–Hopf bifurcation, since that requires $c_1 > 0$. ⌟

2 What remains open

The general question of [1], for all case-9 and case-10 networks without an Andronov–Hopf bifurcation, is not settled here: case 9 outside the family $c_2 = d_2 = -1$, $d_3 = 0$ (in particular $d_2 \neq -1$ or $d_3 \neq 0$, including the branch $c_3 = 0$), and all of case 10, remain open. The same strategy with Dulac functions $x^\alpha y^\beta$ reduces it to sign conditions on a cubic-plus-quadratic form.

3 Computer verification

`scripts/certify_dulac.py` checks every identity above symbolically (sympy): the Dulac computation, the coefficients and discriminant in the subfamily and in the example, the vector $u = c \times d$, the equilibrium and its Jacobian, and the boundary behaviour. The verifier's scripts (`verification/`) recheck these identities with separate code, and also check: the ODEs against the networks via a generic mass-action constructor, the optimality of b, the sign claims, the statements of the remark added at verification, and the specialisation of the Hopf condition of Theorem 9 [1]. A numerical sanity check (not part of the proof) finds no cycles for $d_1 \geq 2c_1$, and finds the predicted small stable limit cycles just beyond the Hopf curve for $c_1 < d_1 < 2c_1$.

References

[1] Murad Banaji, Balázs Boros and Josef Hofbauer. Oscillations in three-reaction quadratic mass-action systems. *Studies in Applied Mathematics*, **2024**.
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